

1. Write a C++ program that prompts the user to enter two values and prints them after swapping the entered values, using swap function (after declaring it). Call the function by 3 methods: a) by value b) by reference c) by address

Solution:

a) Call by Value:

```
#include<iostream>
using namespace std;
void swap(int x, int y)
{
    int temp = x;
    x = y;
    y = temp;
    cout << "Inside the function, the numbers are: " << x << " " << y << endl;
}
void main()
{
    int a, b;
    cout << "Enter 2 numbers: " << endl;
    cin >> a >> b;
    cout << "The numbers before swapping are: " << a << " " << b << endl;
    swap(a, b);
    cout << "Inside the main, the numbers are: " << a << " " << b << endl; // no
change for a,b
}
```

b) Call by Reference:

```
#include<iostream>
using namespace std;
void swap(int &x, int &y)
{
    int temp = x;
    x = y;
    y = temp;
    cout << "Inside the function, the numbers are: " << x << " " << y << endl;
}
void main()
{
    int a, b;
    cout << "Enter 2 numbers: " << endl;
    cin >> a >> b;
    cout << "The numbers before swapping are: " << a << " " << b << endl;
    swap(a, b);
    cout << "Inside the main, the numbers are: " << a << " " << b << endl; // a, b
will also be swapped
}
```

c) Call by Address (using pointers):

```
#include<iostream>
using namespace std;
void swap(int *x, int *y)
```

```

{
    int temp = *x;
    *x = *y;
    *y = temp;
    cout << "Inside the function, the numbers are: " << x << " " << y << endl;
}
void main()
{
    int a, b;
    cout << "Enter 2 numbers: " << endl;
    cin >> a >> b;
    cout << "The numbers before swapping are: " << a << " " << b << endl;
    swap(&a, &b);
    cout << "Inside the main, the numbers are: " << a << " " << b << endl; // a, b
will also be swapped
}

```

2. Write a C++ program that defines a function taking 2 numbers from the user: a double base and an integer power. the function returns the result of raising the double base to the integer power

```

#include<iostream>
using namespace std;
double power (double x, int n)
{
    double result = 1.0;
    for (int i=1; i<=n; i++)
        result = result * x;
    return result;
}
void main( )
{
    double a;
    int b;
    cout<< "Enter a double base then an integer power: "<<endl;
    cin>>a>>b;
    cout<<a << " ^ "<<b<<" = "<< power (a , b) << endl;
}

```

3. Write a C++ program containing a function *factorial* which takes 2 parameters: an array of integers and its size. The function calculates the factorial of each element in the array and prints the result in another array.(Hint: factorial (x [i]) = 1*2*3*...x [i])

Solution:

```

#include <iostream>
using namespace std;
const int n = 3;
void factorial(int x[]);

```

```

void main()
{
    int a[n];
    cout << " Enter " << n << " numbers " << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> a[i];
    }
    factorial(a); // function calling
} // end of main ( )

void factorial(int x[]) {
    int f[n];
    for (int i = 0; i < n; i++) {
        f[i] = 1;
        for (int k = 1; k <= x[i]; k++)
        {
            f[i] = f[i] * k;
        }
        cout << f[i] << endl;
    }
} // end of function

```

4. Write a C++ program containing a function *power* which takes 2 parameters: an array of integers and the power value. The function calculates raise each element in the array to the power value and prints the result in another array.

```

#include <iostream>
using namespace std;
const int n = 3;
void power(int x[], int p);
void main()
{
    int a[n],p;
    cout << " Enter " << n << " numbers " << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> a[i];
    }
    cout << "enter the power value:" << endl;
    cin >> p;
    power(a,p); // function calling
} // end of main ( )

void power(int x[],int p) {
    int r[n];
    for (int i = 0; i < n; i++) {
        r[i] = 1;
        for (int k = 1; k <= p; k++)
        {
            r[i] = r[i] * x[i];
        }
        cout << r[i] << endl;
    }
} // end of function

```

5. Write a C++ program containing a function “maxarr” that reads an array of 10 integer numbers from user then return the maximum number in the array.

```
#include <iostream>
using namespace std;
const int n = 10;
int maxarr(int a[]);
void main()
{
    int x[n];
    int maxno;
    cout << " Enter " << n << " numbers " << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> x[i];
    }
    maxno=maxarr(x); // function calling
    cout << "the maximum no is:" << maxno << endl;
} // end of main ( )

int maxarr(int a[]) {
    int max=a[0];
    for (int i = 0; i < n; i++)
    {
        if( a[i] > max)
        {
            max=a[i] ;
        }
    }
    return max;
} // end of function
```

6. Write a C++ program containing a function “avg” that reads an array of 10 integer numbers from user then return the average of its numbers.

```
#include <iostream>
using namespace std;

const int n = 10;
float avg(int a[]);
void main()
{
    int x[n];
    float average;
    cout << " Enter " << n << " numbers " << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> x[i];
    }
    average=avg(x); // function calling
    cout << "the average is:" << average << endl;
} // end of main ( )

float avg(int a[]) {
    float sum=0;
    for (int i = 0; i < n; i++)
```

```

    {

        sum = sum + a[i];

    }

    return sum/n;
} // end of function

```

7. **Write a C++ program containing a function “sum2arr” that takes 2 arrays from the user then add the elements of the two arrays , store the result in a third array then print the third array**

```

#include <iostream>
using namespace std;

const int n = 5;
void sum2arr(int a[],int b[]);
void main()
{
    int x[n],y[n];
    cout << "enter first array elements" << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> x[i];
    }
    cout << " enter second array elements" << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> y[i];
    }
    sum2arr(x, y);
} // end of main ( )

void sum2arr(int a[],int b[])
{
    int res[n];
    cout << "the sum of the two arrays are:" << endl;
    for (int i = 0; i < n; i++)
    {

        res[i] = a[i] +b[i];

        cout << res[i] << endl;
    }
} // end of function

```

8. **Write a C++ program containing a function “multiply” that takes 2 arrays from the user then multiply the elements of the two arrays , store the result in a third array then print the third array.**

```

#include <iostream>
using namespace std;

const int n = 3;
void multiply(int a[],int b[]);
void main()

```

```

{
    int x[n],y[n];
    cout << "enter first array elements" << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> x[i];
    }
    cout << " enter second array elements" << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> y[i];
    }
    multiply(x, y);
} // end of main ( )

void multiply(int a[],int b[])
{
    int res[n];
    cout << "the sum of the two arrays are:" << endl;
    for (int i = 0; i < n; i++) {
        res[i] = a[i] * b[i];

        cout << res[i] << endl;
    }
} // end of function

```

9. Write a C++ program containing a function “multint” that takes an array of n elements and also an integer number. The function should multiply each element in the array by the number entered by user.

```

#include <iostream>
using namespace std;
const int n = 3;
void multint(int *a,int m);
void main()
{
    int x[n];
    int mult;
    cout << "enter array elements" << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> x[i];
    }
    cout << "enter the value to multiply:" << endl;
    cin >> mult;

    multint(x, mult);
    cout << "the value of the array after multiply:" << endl;
    for (int i = 0; i < n; i++)
    {
        cout << x[i] << endl;
    }
} // end of main ( )

void multint(int *a,int m)

```

```

{

    for (int i = 0; i < n; i++) {

        a[i] = a[i] * m;

    }
} // end of function

```

- 10. Write a C++ program containing a function “decrement” that takes an array of n elements and also an integer number. The function should decrement each element in the array by the number entered by user.**

```

#include <iostream>
using namespace std;
const int n = 3;
void decrement(int *a,int d);
void main()
{
    int x[n];
    int dec;
    cout << "enter array elements" << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> x[i];
    }
    cout << "enter the value to decrement:" << endl;
    cin >> dec;

    decrement(x, dec);
    cout << "the value of the array after decrement:" << endl;
    for (int i = 0; i < n; i++)
    {
        cout << x[i] << endl;
    }
} // end of main ( )

void decrement(int *a,int d)
{

    for (int i = 0; i < n; i++) {

        a[i] = a[i] - d;

    }
} // end of function

```

- 11. (very important) Write a C++ program containing 2 functions: the first is increment which increments each element in the array by a fixed value but not exceed 10, while the second is print which prints the array elements after being incremented. Both functions take a pointer to an array of integers and also its size as parameters,**

Solution:

```

#include <iostream>
using namespace std;

```

```

const int n = 5;
void increment(int *x, int bonus);
void print(int *x);
void main()
{
    int b, x[n];
    cout << "-----" << endl;
    cout << "| Please enter " << n << " numbers |" << endl;
    cout << "-----" << endl;
    for (int i = 0; i <= n - 1; i++) {
        cin >> x[i];
    }
    print(x);
    cout << "-----" << endl;
    cout << "| Enter the value of increment |" << endl;
    cout << "-----" << endl;
    cin >> b;
    increment(x, b);
    print(x);
}
void print(int *x)
{
    for (int i = 0; i <= n - 1; i++)
    {
        cout << x[i] << endl;;
    }
} // end of function print
void increment(int *x, int bonus) {
    for (int i = 0; i <= n - 1; i++)
    {
        if (x[i] + bonus <= 10)
            x[i] = x[i] + bonus;
        else
            x[i] = 10;
    }
} // end of function increment

```

Write a C++ function named "Inc" having two parameters which increment the first parameter with the value of the second parameter. "Using call by reference"

```

using namespace std;
void inc(int &x, int v);
void main()
{
    int a, b;
    cout << "enter a number" << endl;
    cin >> a;
    cout << "enter increment value" << endl;
    cin >> b;
    inc( a, b);
    cout << "the number value after increment=" << a;
}
void inc(int &x, int v)
{
    x = x + v;
}

```

a) Develop a C++ function that takes an array of float numbers, its length and an integer parameter. The functions increment all the array elements with the value of the last parameter.

```
#include <iostream>
using namespace std;
void increment(float *a, int t, int b);
void main()
{
    const int n = 5;
    float x[n];
    int incv;
    cout << "enter array elements" << endl;
    for (int i = 0; i < n; i++)
    {
        cin >> x[i];
    }
    cout << "enter the value to increment:" << endl;
    cin >> incv;
    increment(x, n, incv);
    cout << "the value of the array after increment:" << endl;
    for (int i = 0; i < n; i++)
    {
        cout << x[i] << endl;
    }
}
void increment(float *a, int t, int b)
{
    for (int i = 0; i < t; i++)
    {
        a[i] = a[i] + b;
    }
}
```